

Three Ways to Estimate Kinship Under Population Structure

PC-Relate, PCAngsd and evalAdmix compared, and how PC-Relate derives (k_0, k_1, k_2)

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1 They are the same estimator

The headline result of this note is that PC-Relate, PCAngsd’s kinship estimator and evalAdmix’s correlation of residuals are not three competing ideas. They are one idea, instantiated three times, and they differ only in bookkeeping. The idea is:

Predict each individual’s genotype from their *ancestry* alone. Whatever is left over — the residual — is what ancestry cannot explain. Two individuals who share recent ancestors deviate from their predictions in the *same direction*, so the covariance of their residuals measures relatedness.

Formally, let $g_{is} \in \{0, 1, 2\}$ be the allele count for individual i at site s , and let π_{is} be the *individual allele frequency* — the frequency expected at site s given i ’s ancestry. Because a genotype is the sum of two alleles,

$$\text{Cov}(g_{is}, g_{js}) = 4\phi_{ij} \pi_s(1 - \pi_s), \quad \text{Var}(g_{is}) = 2\pi_s(1 - \pi_s), \quad (1)$$

so the *correlation* of the two residuals is $2\phi_{ij}$ and the covariance normalised by $4\pi(1 - \pi)$ is ϕ_{ij} itself. Every method below is a sample version of Equation (1). What separates them is exactly three choices:

- (i) **Where $\hat{\pi}$ comes from** — a projection onto principal components, or a discrete K -population admixture model.
- (ii) **What goes in the denominator** — the variance the *model* predicts, $\sqrt{\hat{\pi}(1 - \hat{\pi})}$, or the variance the residuals *actually have*.
- (iii) **Whether the pair’s own genotypes were used to build $\hat{\pi}$** — and if so, whether anything is done about it.

Choice (iii) is the one that matters most in practice, and it is where the three methods genuinely diverge.

2 The three methods

2.1 PC-Relate

$\hat{\pi}$ comes from a linear regression of genotypes on ancestry-representative principal components: with V the matrix of PCs (plus an intercept), $2\hat{\mu}_{is}$ is the fitted value of g_{is} . The residual is $R_{is} = g_{is} - 2\hat{\mu}_{is}$ and

$$\hat{\phi}_{ij}^{\text{PC-Relate}} = \frac{\sum_s R_{is} R_{js}}{4 \sum_s \sqrt{\hat{\mu}_{is}(1 - \hat{\mu}_{is})} \hat{\mu}_{js}(1 - \hat{\mu}_{js})}. \quad (2)$$

The denominator is the *model-expected* residual variance. As instructed, we set aside here the fact that GENESIS obtains the PCs from PC-AiR on a mutually unrelated training set; that is a property of how the PCs are built, not of the estimator. PC-Relate is the only one of the three that goes on to produce IBD sharing probabilities (Section 6).

2.2 PCAngsd

$\hat{\pi}$ comes from a truncated SVD of the centred genotype *dosages*, iterated: estimate population frequencies $\hat{p}_s$, form posterior genotype probabilities and dosages $E[g_{is} \mid X_{is}, \hat{\pi}_{is}]$ from genotype likelihoods, take the rank- D SVD of the centred dosages, add $2\hat{p}_s$ back, and feed $\hat{\pi}$ in again as the prior until convergence.

The kinship estimator itself is then *identical in form* to Equation (2). Reading it off the source of the last version that shipped it (`kinship.py`, `kinship_cy.pyx`): the numerator matrix is set to $E - 2\hat{\Pi}$, the denominator matrix to $\sqrt{\hat{\Pi}(1 - \hat{\Pi})}$, and the result is their cross-product ratio divided by 4 — Equation (2) with dosages in place of called genotypes. The file is even named for it: its docstring reads “*Kinship estimator using genotype likelihoods based on PC-Relate*”, and the function is `kinshipConomos`. The diagonal is handled separately, from the genotype posteriors, to give inbreeding coefficients.

So PCAngsd’s kinship is PC-Relate, with two substantive changes: $\hat{\pi}$ is obtained by iterated SVD rather than one regression, and the input is genotype likelihoods, so nothing is ever hard-called. That second point is the real contribution — it is what makes the estimator usable at low depth.

Status. This is no longer available. The `-kinship` flag, and the companion `-relate` option for dropping related individuals from a Beagle file, were removed in the January 2021 Cython rework; `kinship.py` and `kinship_cy.pyx` were deleted the same day. Current PCAngsd (v1.36) exposes only `--inbreed-sites` and `--inbreed-samples`. PCAone, which re-implements the PCAngsd algorithm, likewise offers `--inbreed` but no kinship. For low-depth relatedness today the supported route is NGSremix, using PCAngsd or NGSadmixmap as the front-end.

2.3 evalAdmix

$\hat{\pi}$ comes from the admixture model: $\hat{\pi}_{is} = \sum_k \hat{q}_{ik} \hat{f}_{ks}$, with $\hat{q}$ and $\hat{f}$ from ADMIXTURE, STRUCTURE or NGSadmixmap. The residual is again $r_{is} = g_{is} - 2\hat{\pi}_{is}$, but the statistic is an ordinary **Pearson correlation**:

$$c_{ij} = \frac{\sum_s (r_{is} - \bar{r}_i)(r_{js} - \bar{r}_j)}{\sqrt{\sum_s (r_{is} - \bar{r}_i)^2} \sqrt{\sum_s (r_{js} - \bar{r}_j)^2}} \rightarrow 2\phi_{ij}. \quad (3)$$

Two differences from Equation (2) follow. First, **the scale**: a correlation estimates 2ϕ , not ϕ , so duplicate pairs return 1 and full sibs $\frac{1}{2}$. Second, and more interesting, **the denominator is empirical** — the residuals’ own observed variance, not $\hat{\pi}(1 - \hat{\pi})$. That makes the statistic self-calibrating: anything that inflates residual variance uniformly (genotyping error, a slightly wrong $\hat{\pi}$, unmodelled noise) cancels between numerator and denominator, where in Equation (2) it would bias $\hat{\phi}$ downward.

The leave-one-out correction. This is evalAdmix’s distinctive feature and the answer to choice (iii) above. Before correlating individual i with individual j , it re-estimates the ancestral frequencies F with i excluded (function `adaptF`, an EM update run `-nIts` times, default 5), recomputes j ’s residuals under those frequencies, and correlates i ’s original residual against j ’s adapted residual. The point is that if i ’s own genotypes helped define the frequencies, then i ’s residual is shrunk toward zero and the kinship is attenuated. Removing i from the frequency estimation removes that feedback.

The corrected estimator. Version 1.0 adds `-method corrected`, the analytic estimator of van Waaij et al. (2023): instead of leave-one-out refitting, compute the empirical residual correlation $\hat{b}$ and subtract the correlation $\hat{c}$ that the fitted model itself implies, obtained from $(\mathbf{I} - \mathbf{P})\hat{\mathbf{D}}(\mathbf{I} - \mathbf{P})$ where $\mathbf{P}$ projects onto the column space of Q (or of the PCs). Single-pass instead of iterative, and it needs only genotypes and Q — no .P file. Under a correct model $\hat{b} - \hat{c} \rightarrow 0$ as sites accumulate.

Note on intent. evalAdmix was written to *test admixture model fit*, not to estimate kinship; its documentation mentions relatedness only as a caveat (“positive correlation . . . might also be due to relatedness”). But Equation (3) is a kinship estimator whether or not it was meant as one.

3 Side by side

	PC-Relate	PCAngsd	evalAdmix
$\hat{\pi}$ from	regression on ancestry PCs	iterated truncated SVD on GL dosages	admixture model $\sum_k \hat{q}_{ik} \hat{f}_{ks}$
Input	called genotypes	genotype likelihoods	genotypes or GLs (Beagle)
Needs K ?	no	no	yes
Statistic	Eq. (2)	Eq. (2) (dosages)	Pearson corr., Eq. (3)
Denominator	model, $\sqrt{\hat{\pi}(1 - \hat{\pi})}$	model, $\sqrt{\hat{\pi}(1 - \hat{\pi})}$	empirical residual SD
Returns	ϕ	ϕ	2ϕ
Self in $\hat{\pi}$?	yes, uncorrected	yes, uncorrected	removed (leave-one-out or analytic)
(k_0, k_1, k_2) ?	yes	no	no
Inbreeding	yes	yes	no
Available?	yes (GENESIS)	removed in 2021	yes

4 An empirical comparison

All three were run on the 126-individual, 104,290-SNP example dataset shipped with RelateAdmix ($K = 2$), which is simulated with a known pedigree: ten pairs each of duplicates, parent–offspring, full sibs, half sibs, first cousins and second cousins, and 7,815 unrelated pairs. PC-Relate is the genuine GENESIS `pcrelate()`; evalAdmix is the real program, both of its estimators. RelateAdmix is shown as a reference point — it is a

different kind of method (constrained full ML), not one of the three; see Section 4.4 before reading its RMSE as a ranking. Exact commands and data provenance are in Section 7.

Relationship	true ϕ	PC-Relate (default)	PC-Relate (corr. off)	PCAngsd	evalAdmix	(RelateAdmix)
unrelated ($n=7815$)	0	+0.00002	−0.0055	−0.0055	−0.0016	+0.0004
second cousin ($n=10$)	0.0156	0.0200	0.0132	0.0130	0.0177	0.0116
first cousin ($n=10$)	0.0625	0.0653	0.0564	0.0560	0.0620	0.0526
half sib ($n=10$)	0.125	0.1268	0.1199	0.1185	0.1223	0.1108
full sib ($n=10$)	0.25	0.2470	0.2410	0.2410	0.2464	0.2439
parent–offspring ($n=10$)	0.25	0.2503	0.2390	0.2374	0.2475	0.2500
duplicate ($n=10$)	0.5	0.4972	0.4860	0.4868	0.5000	0.5000
correlation with truth		0.9891	0.9630	0.9628	0.9883	0.9994
RMSE		0.00331	0.00824	0.00825	0.00378	0.00089
SD among unrelated		0.00330	0.00616	0.00617	0.00345	0.00035

4.1 The whole difference is one post-hoc correction

The first run of this comparison used a re-implementation of Equation (2) rather than GENESIS, and produced the “corr. off” column: attenuation of 3–17% and unrelated pairs sitting at −0.0055 instead of 0. The real `pcrelate()` does not do that. Tracking down why turned out to be the most informative result here.

It is not the individual allele frequencies. GENESIS computes $\hat{\mu} = \frac{1}{2}V(V^\top V)^{-1}V^\top G$ (functions `.calcISAFBeta`, `.estISAF`) — an ordinary hat-matrix regression, exactly as re-implemented. Substituting GENESIS’s own PCs changes nothing (unrelated mean −0.0053), and its default `maf.bound.method = "filter"`, which drops the 20.4% of sites where some individual’s fitted $\hat{\mu}$ leaves the MAF range, changes nothing either (−0.0055).

The difference is `correctKin`, enabled by the default `small.samp.correct = TRUE`. For each PC k , it regresses the kinship estimates on the entries of $V_k V_k^\top$, fits that line using *only* pairs with $\hat{\phi} < 2^{-11/2} \approx 0.022$ — i.e. the effectively unrelated ones — and subtracts it from every pair. Running `pcrelate()` with `small.samp.correct = FALSE` reproduces the re-implementation at a correlation of **1.0000**, with the unrelated mean identical to five decimals (−0.00545) and the same RMSE (0.00824).

So the raw estimator of Equation (2) *is* biased downward, for the reason given under choice (iii): the pair’s own genotypes sit inside $\hat{\pi}$, the PCs are partly fitted to the relatedness being measured, and the residuals shrink. PC-Relate does not avoid this — it measures the bias on the unrelated pairs and subtracts it afterwards. The correction is worth a factor of 2.5 in RMSE (0.00824 $\rightarrow$ 0.00331) and it is what makes PC-Relate competitive.

This also settles the status of PCAngsd’s kinship precisely. It implemented Equation (2) and nothing else — no `correctKin`. Its behaviour is therefore the “corr. off” column, which is why that column and PCAngsd’s agree to within 3×10^{-4} . *PCAngsd’s kinship was PC-Relate’s estimator without PC-Relate’s correction.*

4.2 Four ways to run evalAdmix

evalAdmix’s residual-correlation statistic can be reached by four routes, differing in how the mean structure is fitted and how the self-contribution is removed. The fourth is the PCAone `--evaladmix` implementation of Section 5.1; all four were run on the same data.

	ADMIXTURE + EM	ADMIXTURE + projection	PCA ($K-1$) + projection (R)	PCAone --evaladmix
Fits mean structure with	Q, F	Q	$K-1$ PCs + int.	$K-1$ PCs + int.
Removes self by	leave-one-out	$\hat{b} - \hat{c}$	$\hat{b} - \hat{c}$	$\hat{b} - \hat{c}$
Passes over data	-nIts per ind.	single	single	single
Holds genotypes in RAM	yes	yes	yes	no
Clips to $[-1, 1]$	—	no	no	yes
Needs ADMIXTURE	yes	yes	no	no
correlation with truth	0.9883	0.9883	0.9880	0.9878
RMSE, related	0.00292	0.00429	0.00443	0.00251
<i>same, all clipped</i>	<i>0.00292</i>	<i>0.00255</i>	<i>0.00251</i>	<i>0.00251</i>
RMSE, related excl. dup.	0.00320	0.00279	0.00275	0.00275
SD among unrelated	0.00345	0.00349	0.00354	0.00353
mean, unrelated	-0.00156	-0.00154	-0.00155	-0.00155
mean, duplicates	0.5000	0.5079	0.5083	0.5000
<i>attenuation, estimate/truth</i>				
duplicate	1.000	1.016	1.017	1.000
parent-offspring	0.990	0.998	0.999	0.999
full sib	0.985	0.990	0.990	0.990
half sib	0.979	0.981	0.982	0.982
first cousin	0.992	0.994	0.995	0.995
second cousin	1.134	1.135	1.136	1.136
runtime, this dataset	6 s wall 93 s CPU (20 thr.)	22 s wall 21 s CPU (1 thr.)	2.2 s (R, 1 thr.)	0.76 s wall <i>0.12 s of which is the statistic</i>

Accuracy is essentially identical across all four — correlations with truth differ in the fourth decimal. The apparent RMSE spread in the first row is almost entirely the $[-1, 1]$ clip: apply it uniformly (second row) and the three projection routes converge on 0.00251–0.00255, with the EM route at 0.00292. Excluding duplicates entirely, the projection routes sit at 0.00275–0.00279 against EM’s 0.00320.

Where they genuinely differ is small and systematic. The EM route is exactly right on duplicates by construction and is the tightest on unrelated pairs ($SD = 0.00345$); the projection routes are slightly better on every other class. All four overshoot identically at second cousins (≈ 1.135), confirming that is a property of the statistic rather than of any front-end.

PCAone --evaladmix combines the two advantages: projection accuracy on the non-duplicate classes together with exact duplicates, because it applies the clip that the other projection routes leave to the user. That gives it the lowest as-shipped RMSE of the four (0.00251), though as the clipped row shows, this is a default rather than a better estimator.

The real differences are cost and dependencies. The EM route refits F with each individual excluded, -nIts times, so its work grows with sample size on top of the per-pair cost; the projection routes are single-pass. The two PCA routes need no ADMIXTURE run at all — invisible here, because the .P/.Q files ship with the data, but usually the dominant cost of a real analysis. And only the PCAone implementation avoids holding the genotype matrix in RAM, which is what makes the approach usable at biobank scale; on this dataset the statistic itself costs 0.12s on top of a PCA that was going to be run anyway.

Recommendation: `PCAone -b plink -k <K-1> --evaladmix`. It needs no admixture model, adds almost nothing to a PCA, scales out-of-core, and matches or beats the alternatives on every relationship class except the second-cousin overshoot that all four share.

4.3 Improving the PCA route

If the PCA + projection route is the one you want, four knobs are worth knowing about. Only two of them matter, and one matters enormously.

1. Use exactly $K - 1$ PCs. This dominates everything else. Scanning the number of PCs, with all else fixed:

PCs used	$k = 1 (= K-1)$	$k = 2$	$k = 3$	$k = 4$
RMSE (related pairs)	0.00443	0.03035	0.05001	0.06917
mean at duplicates ($\phi = 0.5$)	0.508	0.541	0.579	0.620
correlation with truth	0.988	0.947	0.891	0.830

One PC too many costs a factor of seven in RMSE. The reason is that with $K = 2$ genuinely distinct populations there is only one real ancestry axis; PC2 onward are fitting *relatedness*, not ancestry, and projecting them out distorts the residual structure the estimator depends on. What actually has to match is the *dimension* of the projection: `method="standard"` appends an intercept row, so $K - 1$ PCs span a K -dimensional

space, the same as Q . Chen & Storey normalisation (`method="CS"`) has no intercept row and therefore needs $k = K$; run at $k = K - 1$ it collapses (RMSE 0.0172, $r = 0.67$). Get the dimension right and the normalisation label barely matters; get it wrong and nothing else can save you.

2. Cap at the parameter bound. A correlation cannot exceed 1, so $\phi \leq 0.5$; but $\hat{b} - \hat{c}$ is unconstrained and overshoots, returning 0.508 for duplicates. Truncating at the bound is as principled as `RelateAdmix`'s simplex constraint and costs nothing:

	RMSE, related	RMSE, related <i>excluding</i> duplicates	duplicates
PCA + projection	0.00443	0.00275	0.5083
+ cap at $\phi = 0.5$	0.00251	0.00275	0.5000
PCA on unrelated subset	0.00415	0.00269	0.5077
+ cap	0.00246	0.00269	0.5000
<i>evalAdmix</i> , <i>ADMIXTURE</i> + <i>EM</i>	0.00292	0.00320	0.5000
<i>PC-Relate</i>	0.00358	0.00353	0.4972

Note the second column carefully. **Excluding duplicates, the untuned PCA route was already the best method in this comparison** (0.00275, against `evalAdmix`'s 0.00320 and `PC-Relate`'s 0.00353). Its entire apparent deficit in Section 4.2 was ten duplicate pairs breaking a bound. Capping is free and correct, but it changes nothing unless duplicates are present — and duplicates are usually removed from real analyses anyway.

3. Compute the PCs on an unrelated subset and project the rest in. This is the PC-AiR idea applied to `evalAdmix`. The analytic $\hat{b} - \hat{c}$ correction handles the *geometry* of the projection, but not the fact that the PCs themselves were fitted partly to the relatedness being measured. Restricting the eigen-decomposition to a maximal unrelated set (66 of 126 individuals here) and projecting everyone onto those loadings improves every class slightly. Judged on the 50 related non-duplicate pairs, where the cap plays no role, RMSE goes from 0.00275 to 0.00269 — **2.3%**. That gain is statistically real (paired t on squared errors $p = 0.013$; bootstrap 95% CI for the RMSE difference $[2 \times 10^{-5}, 1.1 \times 10^{-4}]$, excluding zero; the unrelated-subset version wins in 99.9% of 10,000 bootstrap resamples) but it is practically negligible. And this dataset is an extreme case, with nearly half the sample sitting in a related pair; expect less in a cohort with few relatives. In practice you would iterate: run once, identify relatives, recompute the PCs without them. Worth doing if it is cheap; not worth restructuring a pipeline for.

4. Normalisation and MAF threshold: do not bother. Centring only versus centring and scaling versus Chen & Storey (each at its correct dimension) give RMSE 0.00443, 0.00441 and 0.00444; MAF thresholds of 0.01 and 0.05 are indistinguishable. These are 1% effects.

What is left — and the honest summary. After all of this the PCA route still overshoots at second cousins by $\approx 13\%$, and so do both *ADMIXTURE* routes by the same amount. That is a property of the statistic where the signal approaches the noise floor, not of the front-end, and none of these knobs touches it.

So no better approach was found. Of the four knobs, one (the number of PCs) is a guardrail rather than an improvement — it can only make things much worse if set wrong; two (normalisation, MAF) do nothing; the cap corrects a bound violation that affects duplicates alone; and the unrelated-subset PCA buys 2.3%. Run at $K - 1$ PCs, **PCA + projection was already the most accurate method in this entire comparison** once duplicates are set aside (0.00275, against `evalAdmix` EM's 0.00320, `PC-Relate`'s 0.00353 and `RelateAdmix`'s 0.00812). The right conclusion is not that it needs improving, but that it was already winning and the earlier tables obscured that.

Recommended recipe.

- Choose K ; use $K - 1$ PCs with `method="standard"` (or K with `"CS"`).
- Run once on all individuals; identify related pairs; recompute the PCs on an unrelated subset and project everyone in; re-run.
- Truncate the resulting correlations to $[-1, 1]$ before halving to the ϕ scale.
- Do not spend time on normalisation or MAF thresholds.

4.4 What each method gets right

Expressed as estimate/truth:

	duplicate	parent-offs.	full sib	half sib	1st cousin	2nd cousin
PC-Relate (default)	0.994	1.001	0.988	1.014	1.044	1.283
PC-Relate (corr. off) / PCAngsd	0.972	0.956	0.964	0.959	0.903	0.847
evalAdmix	1.000	0.990	0.985	0.979	0.992	1.134
RelateAdmix	1.000	1.000	0.976	0.886	0.842	0.740

PC-Relate and evalAdmix are both close to unbiased down to first cousins and both overshoot at second cousins, where the signal approaches the noise floor. The uncorrected estimator attenuates monotonically with distance. RelateAdmix shows the opposite pattern: exact for the closest classes, and attenuating hardest for the most distant.

4.5 A warning about overall RMSE — and about RelateAdmix

The RMSE column in the main table is misleading, and the way it misleads is worth spelling out. **99.24% of the 7,875 pairs are unrelated**, so an RMSE computed over all pairs is very nearly a measurement of performance on unrelated pairs alone. Splitting it:

	RMSE, all (7875 pairs)	RMSE, unrelated (7815)	RMSE, related (60)	bias, related
RelateAdmix	0.00089	0.00054	0.00812	−0.00571
PC-Relate	0.00331	0.00330	0.00358	+0.00055
evalAdmix (EM)	0.00378	0.00378	0.00292	−0.00120
PCA + projection	0.00387	0.00386	0.00443	+0.00084
PCAone --evaladmix	0.00385	0.00386	0.00251	−0.00055
PC-Relate (corr. off)	0.00824	0.00823	0.00989	−0.00842

The ranking inverts. On the 60 pairs that actually have relatedness to estimate, PCAone’s --evaladmix is best (0.00251), then evalAdmix’s EM route (0.00292), then PC-Relate (0.00358) — and **RelateAdmix is the worst** (0.00812), more than three times the best. The two evalAdmix rows differ only in the $[-1, 1]$ clip and the front-end; see Section 4.2. Per class it is the worst method at half sibs (0.0152) and first cousins (0.0100), consistent with the attenuation ratios of 0.886 and 0.842 above.

Why its unrelated RMSE is so small is a constraint, not precision. RelateAdmix is a constrained ML estimator: (k_0, k_1, k_2) must lie on the simplex, so $\phi \geq 0$ by construction. Among the 7,815 unrelated pairs it never returns a negative value (range $[0, 0.00384]$) and 93.6% have $\hat{k}_1 < 10^{-3}$ — the estimates are pinned at the boundary. PC-Relate, an unconstrained moment-style estimator, scatters symmetrically about zero (range $[-0.0165, +0.0120]$). Pinning at the truth is not the same as estimating it well; it happens to be right here because the truth for those pairs *is* the boundary.

The same effect explains RelateAdmix’s exactness at the other two boundary classes: duplicates ($k_2 = 1$) and parent-offspring ($k_1 = 1$) are simplex vertices, and it returns 0.00000 RMSE at both. Its errors are concentrated entirely in the interior — full sibs, half sibs, cousins — where no constraint helps.

What this means in practice. RelateAdmix is excellent at the question “which pairs are related at all?” — essentially no false positives, and exact on duplicates and parent-offspring. It is the *least* accurate of these methods at the question “how related are these two?” for anything between full sibs and second cousins. Which one you care about should decide the method, and an overall RMSE will not tell you.

4.6 And the IBD probabilities

PC-Relate’s (k_0, k_1, k_2) , which neither of the other two produces, recover the pedigree cleanly:

Relationship	estimated			expected		
	$\hat{k}_0$	$\hat{k}_1$	$\hat{k}_2$	k_0	k_1	k_2
duplicate	0.000	0.021	0.979	0	0	1
parent-offspring	0.000	1.000	0.000	0	1	0
full sib	0.255	0.502	0.243	0.25	0.50	0.25
half sib	0.495	0.502	0.002	0.50	0.50	0
first cousin	0.738	0.263	−0.001	0.75	0.25	0

This is the concrete payoff of Section 6. Parent–offspring and full sibs both have $\phi = 1/4$, and indeed PC-Relate’s kinship gives 0.2503 and 0.2470 — indistinguishable. The IBD probabilities separate them completely: $k_0 = 0.000$, $k_1 = 1.000$ versus $k_0 = 0.255$, $k_1 = 0.502$, $k_2 = 0.243$. No kinship-only method can do this.

5 Which one is best?

There is no single ordering, and any claim of one should be treated with suspicion — including the obvious one. PC-Relate has the lower *overall* RMSE (0.00331 vs evalAdmix’s 0.00378), but that is the statistic Section 4.4 warns against: 99.24% of pairs are unrelated, so it mostly measures performance on unrelated pairs. Split it and the two swap places depending on the question:

	PC-Relate	evalAdmix (EM)	PCAone --evaladmix
RMSE on unrelated pairs (7815)	0.00330	0.00378	0.00386
RMSE on related pairs (60)	0.00358	0.00292	0.00251
returns (k_0, k_1, k_2)	yes	no	no
dimensionality to choose	# PCs	K	$K-1$ PCs
assumes <i>discrete</i> ancestry	no	yes	no
needs an ADMIXTURE run	no	yes	no
holds genotypes in RAM	yes	yes	no

PC-Relate is better at “*are these two unrelated?*”; the two evalAdmix routes are better at “*how related are they?*”, with the PCAone implementation best of all on related pairs. No margin is large — all three correlate with the truth at 0.988 — and none is the deciding factor in practice.

What actually separates them is not accuracy at all, but the lower half of that table. PC-Relate is the only one returning (k_0, k_1, k_2) and inbreeding coefficients. PCAone’s --evaladmix is the only one that needs no admixture run and never holds the genotype matrix in RAM, which is what decides whether the analysis is possible at all at biobank scale. PCAngsd’s kinship no longer exists. These are categorical differences, and they should drive the choice where a difference in the fourth decimal of an RMSE should not.

PC-Relate does not escape the dimensionality choice. It is tempting to say PC-Relate “needs no K ”, and in one sense it does not: PCs do not assume discrete ancestral populations, so continuous ancestry is handled without pretending otherwise. But the number of PCs entering the regression still has to be chosen, and that is the same decision wearing different clothes — with the same sensitivity. Running `pcrelate()` on these data with one to four PCs, where $K = 2$ makes one correct:

PCs used	1 (correct)	2	3	4
PC-Relate, RMSE on related pairs	0.00358	0.03474	0.06011	0.09459
mean at duplicates ($\phi = 0.5$)	0.4972	0.5395	0.5880	0.6561
correlation with truth	0.9891	0.9747	0.9535	0.9319
<i>PCA + projection, for comparison</i>	<i>0.00443</i>	<i>0.03035</i>	<i>0.05001</i>	<i>0.06917</i>

One PC too many costs a factor of ten, and PC-Relate degrades slightly *faster* than the projection route it is being compared against. Notably the damage is concentrated in the closest pairs — duplicates inflate to 0.66 while parent–offspring stays at 0.251 and first cousins at 0.066 — so it will not announce itself in a summary statistic. The real advantage of the PC route is the discreteness assumption it avoids, not a decision it saves you. A decision rule:

- **Called genotypes, general use: PC-Relate.** Not because it is the most accurate — it is not, on related pairs — but because it is complete: inbreeding coefficients, and the only IBD probabilities on offer. Keep `small.samp.correct = TRUE` — it is the default, it is doing most of the work, and it requires all samples in one block.
- **You need to tell parent–offspring from full sibs: PC-Relate, necessarily.** $\phi = 1/4$ for both; only k_0 separates them.
- **evalAdmix** is a close second on ϕ (and the best of all on duplicates, exactly 0.5), needs no unrelated set and no leverage correction because leave-one-out removes the problem at source rather than after the fact. Use it when you have a defensible K , when you want the model-fit diagnostic in the same run, or as a cross-check on PC-Relate. Remember it returns 2ϕ .

- **Large samples, or you would rather not run ADMIXTURE: PCAone --evaladmix** (Section 5). Same statistic as the evalAdmix projection route, but the mean structure comes from $K - 1$ PCs, and the single-pass formulation means the genotype matrix never has to fit in RAM. It was the most accurate on related pairs here (0.00251), it costs 0.12s on top of a PCA you were running anyway, and it applies the $\phi \leq 0.5$ truncation by default. Like the other evalAdmix routes it returns 2ϕ and no IBD probabilities.
- **Ancestry not discrete: PC-Relate or the PCA route**, neither of which assumes discrete source populations. This does *not* excuse you from choosing the number of axes — see above.
- **Low depth: none of the three.** PCAngsd’s kinship is gone, and as shown it lacked the correction that makes PC-Relate work. Use PCAngsd or NGSadmixture for $\hat{\pi} / \hat{q}, \hat{f}$, then NGSremix — ML, genotype likelihoods, and it returns (k_0, k_1, k_2) .
- **For screening — “is this pair related at all?” — RelateAdmix.** Its constrained ML never returns a negative ϕ and pins unrelated pairs at zero (RMSE 0.00054 there), so false positives are essentially absent, and it is exact on duplicates and parent–offspring. But see Section 4.4: on the related pairs themselves it is the *worst* of these methods (RMSE 0.00812 vs evalAdmix’s 0.00292), badly attenuated at half sibs and cousins. Do not read its low overall RMSE as accuracy.
- **Trap:** if you run evalAdmix to choose K , close relatives appear as model misfit. Remove relatives first, or read the outliers as relatedness.

Scope. This dataset has discrete, well-separated ancestral populations with K known to be correct, $n = 126$, and no missing data. Small n is exactly the regime `correctKin` was written for, so its contribution would be smaller in a biobank; and where ancestry is continuous or K is wrong, $\hat{\pi}$ from the admixture model degrades while $\hat{\pi}$ from PCs does not.

6 How PC-Relate computes (k_0, k_1, k_2)

Since PC-Relate is the only one of the three that does this, it is worth spelling out. What follows is the GENESIS implementation (`pcrelate, ibd.probs = TRUE`), read from source.

k_2 from a dominance coding. Define

$$D_{is} = \begin{cases} \hat{\mu}_{is} & g_{is} = 0 \\ 0 & g_{is} = 1 \\ 1 - \hat{\mu}_{is} & g_{is} = 2 \end{cases} \quad \tilde{D}_{is} = D_{is} - \hat{\mu}_{is}(1 - \hat{\mu}_{is}),$$

so that $E[\tilde{D}_{is}] = 0$ under HWE. The dominance covariance between two individuals is proportional to k_2 alone — it vanishes for IBD-1 sharing — so

$$\hat{k}_2 = \frac{\sum_s \tilde{D}_{is} \tilde{D}_{js}}{\sum_s \hat{\mu}_{is}(1 - \hat{\mu}_{is}) \hat{\mu}_{js}(1 - \hat{\mu}_{js})}.$$

k_0 from opposing homozygotes. If a pair shares *any* allele IBD at a site they cannot be opposing homozygotes. So observed IBS0, over its expectation if the pair shared nothing, estimates k_0 :

$$\hat{k}_0 = \frac{\sum_s [\mathbf{1}(g_{is}=2, g_{js}=0) + \mathbf{1}(g_{is}=0, g_{js}=2)]}{\sum_s [\hat{\mu}_{is}^2(1 - \hat{\mu}_{js})^2 + (1 - \hat{\mu}_{is})^2 \hat{\mu}_{js}^2]}.$$

k_1 by subtraction. $\hat{k}_1 = 1 - \hat{k}_0 - \hat{k}_2$. GENESIS stores `kin`, `k0` and `k2`; k_1 is left as the remainder.

So it does not go from kinship to IBD probabilities. $\hat{\phi}$, $\hat{k}_0$ and $\hat{k}_2$ are three separate statistics computed from the same residuals, and the identity $\phi = k_1/4 + k_2/2$ is not used to derive them. That is deliberate: it is precisely because ϕ cannot distinguish parent–offspring from full sibs that k_0 needs its own estimator, and the IBS0 count is what supplies it ($k_0 = 0$ versus $k_0 = 1/4$).

Three corrections, applied only when IBD probabilities are requested.

- *Inbreeding.* HWE departure inflates the dominance covariance, so the product of the two individuals’ inbreeding coefficients is subtracted: $\hat{k}_2 \leftarrow \hat{k}_2 - \hat{f}_i \hat{f}_j$.
- *Residual ancestry and kinship bias in $\hat{k}_2$.* For each PC $k \geq 2$, regress the current $\hat{k}_2$ on the entries of $V_k V_k^\top$ with linear and quadratic terms, fitting only on pairs with $\hat{\phi} < 2^{-11/2} \approx 0.022$, and subtract the fit. Then regress on $\hat{\phi}$ itself, fitting on pairs with $\hat{k}_2 < 2^{-9/2} \approx 0.044$, and subtract that too.

- k_0 for distant pairs — where the kinship identity finally enters. For pairs below the first-degree boundary, $\hat{\phi} < 2^{-5/2} \approx 0.177$, the IBS0 estimate is discarded and replaced by

$$\hat{k}_0 = 1 - 4\hat{\phi} + \hat{k}_2,$$

which is $\phi = k_1/4 + k_2/2$ with $k_1 = 1 - k_0 - k_2$, rearranged. In distant pairs the IBS0 count is dominated by noise and genotyping error while $\hat{\phi}$ remains well estimated; in first-degree pairs the IBS0 estimator is kept, for the reason above.

7 Running this with PCAone

7.1 Is there an implementation? No.

PCAone has no `evalAdmix`, model-fit or kinship feature — its only relatedness-adjacent output is `--inbreed`. PCAngsd’s kinship was removed in 2021. The only existing implementations of the residual-correlation statistic are the `evalAdmix` C++ program, which projects onto Q and so needs an admixture run, and the `evalPCA()` R code in `popgenDK/evalPopStructure`, which handles PCA but holds the whole genotype matrix in memory.

7.2 But PCAone already computes the expensive part

PCAone’s `-D/--ld` flag, added for ancestry-adjusted LD, writes a `.residuals` file that is precisely G minus its rank- k fit (Data.cpp: `G -= U * S.asDiagonal() * VT`) — the residual matrix the estimator needs. Two things had to be checked, and both hold: the residuals are written on the *raw* genotype scale, agreeing with residuals computed from unstandardised genotypes to 1.2×10^{-7} and not with standardised ones (which would differ by 4.5×10^{-3}); and the rank- k fit of column-centred data spans the same K dimensions as $k = K - 1$ PCs plus an intercept, so `-k <K-1>` is the right setting.

Post-hoc recipe. Run PCAone, then compute $\hat{b} - \hat{c}$ from its outputs:

```
PCAone -b plink -k <K-1> -d 0 -D --maf 0.05 -o out
Rscript evalAdmix_pcaone.R out plink <K-1>
```

Reading `out.residuals` (uint32 M , uint32 N , then $M \times N$ float32, SNP-major) and `out.eigvecs`:

$$\hat{b} = \text{cor}(R), \quad \hat{c} = \text{cov2cor}((\mathbf{I} - \mathbf{P})\hat{\mathbf{D}}(\mathbf{I} - \mathbf{P})), \quad \hat{\phi} = (\hat{b} - \hat{c})/2,$$

with $\mathbf{P}$ the projection onto $[\text{PC}_1 \dots \text{PC}_{K-1}, \mathbf{1}]$ and $\hat{\mathbf{D}}$ the diagonal of per-individual mean heterozygosity $g(2 - g)$. This reproduces `evalPCA()` at $r = 1.00000$ with a maximum absolute difference of 5.9×10^{-8} — float precision. Total cost beyond the PCA: one pass to read the residual file.

7.3 Implemented: `--evaladmix` in PCAone

The algorithm above is implemented in PCAone as `--evaladmix` (`src/EvalAdmix.cpp`, `src/EvalAdmix.hpp`, wired through `Cmd` and `Main`) and is published at github.com/aalbrechtsen/PCAone, together with the three bug fixes described below:

```
PCAone -b plink -k <K-1> --evaladmix --maf 0.05 -o out
```

It writes `out.corres` (the $N \times N$ correlation of residuals) and `out.kinship` (`corres/2`), both with sample IDs from the `.fam`. `--evaladmix-k` selects how many of the computed PCs enter the projection, so one run can produce many PCs for other purposes while the statistic uses $K - 1$.

Validation. Against the `evalPCA()` reference on the `RelateAdmix` test data: $r = 0.999955$, and identical RMSE on related non-duplicate pairs (0.00275 both). The only entries differing by more than 1.6×10^{-4} are the ten duplicate pairs, where the implementation applies the $[-1, 1]$ clip of Section 4.3 and the R reference does not. In-core and out-of-core runs, and `--evaladmix-k 1` against a `-k 1` run, agree exactly (max difference 0). Recovered means: duplicates 0.5000, parent–offspring 0.2496, full sibs 0.2475, half sibs 0.1227, first cousins 0.0622, unrelated -0.0016 .

Implementation notes. Three things needed care.

- *Genotype scale.* PCAone codes genotypes as $\{0, \frac{1}{2}, 1\}$ (BED2GEN0), i.e. $g/2$. Because $\hat{b}$ and $\hat{c}$ are each separately converted to correlations, the constant factor cancels and no rescaling is needed.
- *Centring.* The out-of-core block reader always returns centred genotypes, ignoring `params.center`. This is harmless for A and $\bar{g}$ — per-site centring subtracts $c_s \mathbf{1}$ from column s , and every resulting term carries a factor $(\mathbf{I} - \mathbf{P})\mathbf{1} = \mathbf{0}$ because the projection contains the intercept — but it does corrupt the heterozygosity, which is therefore recovered as $x = G_{\text{centred}} + \hat{f}_s$. The in-core path runs with `center = false` and needs no correction.
- *The PC scores could not be read correctly.* `Utils::read_usv()` was returning a transposed matrix for any file with more than one column — see below. It is now fixed, and this module uses it.

Limitations. PLINK/PGEN input only; `--maf` with out-of-core is rejected by PCAone itself, unrelated to this feature; and complete data is assumed, as above. Plain PCA and `-D` runs are unaffected.

7.4 Three bugs found along the way

Building the feature surfaced three pre-existing defects in PCAone, all now fixed and documented in the repository’s `docs/` directory. The first is much the most consequential.

1. `read_usv()` transposes any multi-column matrix. The parser fills a flat buffer row by row, then returned `Eigen::Map<Mat2D>(V.data(), j, k)` — and `Mat2D` is column-major. Mapping a row-major buffer as column-major scrambles the contents, equivalent to reading the transpose. It is correct only when there is one column, which is why it went unnoticed.

The victim is not this module but **the ancestry-adjusted LD path**, `LD.cpp:493`, which builds the projector $UU^\top$ from the result. With $k > 1$ the residuals — and therefore every R^2 , pruning and clumping result — were computed against the wrong subspace. Building both versions and running the affected path on the test data with $K = 2$:

finite SNP pairs compared	4,611,432
R^2 values that differ	4,609,439 (99.96%)
correlation between old and new R^2	0.694
mean R^2	0.01240 $\rightarrow$ 0.01014
largest single change	0.2556 $\rightarrow$ 0.0329
pairs with $R^2 > 0.2$	370 (old) $\rightarrow$ 58 (new)
flagged by old but not new	322

The old code systematically inflates LD, and at `--ld-r2 0.2` would prune roughly six times as many pairs as it should. R^2 recomputed independently in R for the six most-affected pairs agrees with the fixed version exactly in every case, confirming which side is correct.

2. Zero-variance variants give NaN R^2 . The LD routines begin from `1/calc_sds(G)`. A monomorphic variant, or one lying entirely in the span of the PCs being adjusted for, has `sd = 0`, so this is ∞ ; the correlation $0 \cdot \infty$ is NaN, written to `.ld.gz` for every pair involving that variant. `calc_sds()` already carried the comment “*return 1e-9 when sd is 0*” — the case was anticipated but the guard never written. On the test data 4 monomorphic variants of 104,290 produced 244 `-nan` rows. Zero-variance columns now yield $R^2 = 0$, which is what is true of a variant carrying no information, with a warning naming the count; all 4,611,432 other values are bit-identical.

3. A missing `.mbim` segfaults. `read_frq()` opened its file without checking, so a missing one read as empty, `F` stayed size zero, and the first `F(snp_idx)` indexed out of bounds — exit 139, no message. This is easy to reach because `-P/--USV` points `filebim` at `<prefix>.mbim` and *only* a run with `-D/--ld` writes one, so a prefix from a plain PCA run looks valid and crashes. It now names the missing file and exits cleanly.

7.5 The derivation: why no residual file is needed

A direct implementation should not write residuals at all. Because $R = G(\mathbf{I} - \mathbf{P})$ and `cor` centres each individual’s residuals across sites,

$$\tilde{R}^\top \tilde{R} = (\mathbf{I} - \mathbf{P})[G^\top G - M \bar{g} \bar{g}^\top](\mathbf{I} - \mathbf{P}), \quad \hat{b} = \text{cov2cor}(\tilde{R}^\top \tilde{R}), \quad (4)$$

so the statistic depends on the data only through three quantities:

- $G^\top G$, the $N \times N$ genotype Gram matrix;
- $\bar{g}$, the per-individual mean genotype;
- $\hat{\mathbf{D}} = \text{diag}(\bar{g}(2 - \bar{g}))$, per-individual mean heterozygosity.

All three are simple accumulations over sites, computable in the *same streaming pass* PCAone already makes, with no second pass, no residual file and no need to hold genotypes. Everything after the pass is $O(N^3)$ on $N \times N$ matrices. Verified numerically: Equation (4) reproduces the direct residual computation to 10^{-15} .

This is a natural fit for PCAone’s out-of-core design, and the marginal cost over a PCA it is doing anyway is one $O(MN^2)$ accumulation — the same order as forming the covariance matrix it already builds. A reasonable interface would be a `--evaladmix` flag emitting the $N \times N$ correlation-of-residuals matrix, with k defaulting to $K - 1$.

Complete data assumed. Everything in this section assumes no missing genotypes, which is the case for the data used here. That assumption is what makes the single-pass form exact: with missingness, evalAdmix skips sites absent in *either* member of a pair, so each pair sees a different set of sites, $G^\top G$ ceases to be a sufficient summary, and one would have to accumulate pairwise site counts alongside it (or mean-impute, at the cost of shrinking the residuals). With complete data no such bookkeeping is needed and Equation (4) holds exactly.

The algorithm in full. Given complete genotypes, K , and $k = K - 1$ PCs:

One streaming pass over sites $s = 1 \dots M$, accumulating three objects:

$$\begin{aligned} A &+= g_s g_s^\top && (N \times N; \text{ this is } G^\top G) \\ b &+= g_s && (N; \text{ gives } \bar{g} = b/M) \\ d &+= g_s \odot (2 - g_s) && (N; \text{ gives } \hat{\mathbf{D}} = \text{diag}(d/M)) \end{aligned}$$

Then, once:

$$\begin{aligned} \mathbf{P} &= V(V^\top V)^{-1} V^\top, \quad V = [\text{PC}_1 \dots \text{PC}_k, \mathbf{1}] \\ \hat{b} &= \text{cov2cor}((\mathbf{I} - \mathbf{P})[A - M\bar{g}\bar{g}^\top](\mathbf{I} - \mathbf{P})) \\ \hat{c} &= \text{cov2cor}((\mathbf{I} - \mathbf{P})\hat{\mathbf{D}}(\mathbf{I} - \mathbf{P})) \\ \hat{\phi} &= \frac{1}{2} \text{clip}(\hat{b} - \hat{c}, -1, 1) \end{aligned}$$

The pass is $O(MN^2)$ — the same order as the covariance matrix PCAone already forms — and the tail is $O(N^3)$. Memory is $O(N^2)$, independent of the number of sites, so nothing about it conflicts with out-of-core operation. Two points of care remain: $\hat{\mathbf{D}}$ assumes Hardy–Weinberg within individuals, so inbred samples need attention; and the $\phi \leq 0.5$ truncation of Section 4.3 belongs on the output.

8 Methods: data and commands

8.1 Data

No data were simulated for this note. The dataset is the example distributed *with* RelateAdmix, in its `data/` directory, obtained by cloning the repository:

```
git clone https://github.com/aalbrechtsen/relateAdmix.git
```

It consists of `smallPlink.{bed,bim,fam}` — 126 individuals, 104,290 autosomal SNPs, no missing genotypes — together with `smallPlink.2.P` (ancestral allele frequencies) and `smallPlink.2.Q` (admixture proportions), which are ADMIXTURE output for $K = 2$ shipped alongside. Because those two files are provided, ADMIXTURE itself was never run; every method that needs $\hat{q}, \hat{f}$ used the same supplied files, so no method is advantaged by a better admixture fit.

The individuals are simulated from two source populations with a known pedigree. The design is visible in the data — individuals are ordered so that relatives are consecutive even–odd pairs — and was read off directly: individuals 0–5 unrelated, 6–25 duplicates, 26–45 parent–offspring, 46–65 full sibs, 66–85 half sibs, 86–105 first cousins, 106–125 second cousins, ten pairs per class. “True ϕ ” throughout is the *theoretical* kinship for that relationship ($\frac{1}{2}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{64}$, and 0 for the remaining 7,815 pairs), not any method’s output.

8.2 Software and commands

RelateAdmix (commit as cloned above) — C++ command-line build, `cp CPP_Makefile Makefile && make`:

```
./relateAdmix -plink smallPlink -f smallPlink.2.P -q smallPlink.2.Q -P 20
```

5.4s wall, 100s CPU on 20 threads; output `output.k` with (k_0, k_1, k_2) per pair.

evalAdmix v1.0, cloned from github.com/GenisGE/evalAdmix and built with `make`. Both estimators were run, on the same PLINK files and the same `.P/.Q`:

```
# default: leave-one-out frequency correction (-nIts 5)
./evalAdmix -plink smallPlink -fname smallPlink.2.P -qname smallPlink.2.Q -P 20 -o corres.txt

# van Waaij et al. analytic projection estimator
./evalAdmix -plink smallPlink -fname smallPlink.2.P -qname smallPlink.2.Q -method corrected \
-P 20 -o corres_corrected.txt
```

Default: 6s wall / 93s CPU (parallel over 20 threads). Corrected: 22s wall / 21.5s CPU — fewer operations in total, single pass rather than `-nIts` refits, but effectively single-threaded, so it is slower in wall time at this sample size and would win as n grows. Both write a 126×126 matrix of correlations of residuals; these were divided by 2 to put them on the kinship scale. The two agreed closely (RMSE 0.00378 vs 0.00382), so the default estimator is quoted in the main table and the corrected one reported alongside.

evalAdmix, third route (PCA + projection) — the R implementation from github.com/popgenDK/evalPopStructure (R/evalPCA.R), which applies the same $\hat{b} - \hat{c}$ estimator to structure from PCA or from the admixture model:

```
pca <- makePCA(g, method="standard", center=TRUE, scale=FALSE)
ePCA <- evalPCA(pca, k = K-1)
eADM <- evalPCA(list(method="admixture", g=g, Q=Q), k = K)
```

with `g` the $102,185 \times 126$ SNP-by-individual genotype matrix. The admixture call reproduces the compiled `-method corrected` output at $r = 1.00000$, confirming the two implementations agree.

PC-Relate — GENESIS 2.32.0, installed from Bioconductor (`BiocManager::install("GENESIS")`), pulling `SNPRelate`, `GWASTools`, `gdsfmt`, `SeqArray`, `SeqVarTools`). The PLINK files were converted with `SNPRelate::snpgdsBED2GL`; ancestry PCs came from `snpgdsPCA(maf=0.05, missing.rate=0.05)`; then

```
pcrelate(GenotypeBlockIterator(geno, snpBlock=20000),
pcs = pcs[,1,drop=FALSE], ibd.probs = TRUE, maf.thresh = 0.05)
```

8.5s. One PC was used, matching $K = 2$. Per the brief, PC-AiR was *not* used — the PCs come from plain PCA on all 126 individuals, so no unrelated training set is involved (`training.set = NULL`). The run was repeated with `small.samp.correct = FALSE` to isolate the contribution of `correctKin`.

PCAngsd — *could not be run*, because the kinship estimator no longer exists: the `-kinship` flag and `kinship.py/kinship_cy.pyx` were removed in January 2021 (commit `3fda6ed`), and current v1.36 offers only inbreeding. Its formula was therefore read from the last version that shipped it (`git show a50c6af^:kinship.py`) and confirmed to be Equation (2). The “PCAngsd” column is that formula applied to these genotypes with $\hat{\pi}$ from a rank-1 SVD of the centred dosages — which, as Section 4.1 shows, coincides with PC-Relate with the correction disabled.

8.3 Code availability

The `--evaladmix` implementation and the three fixes above are at github.com/aalbrechtsen/PCAone (branch `main`), with `docs/evaladmix.md`, `docs/read-usv-fix.md` and `docs/ld-robustness-fixes.md` covering them. A clean clone builds with `make` and reproduces every number in Section 4 bit-identically.

8.4 Reproducibility notes

Two columns are re-implementations rather than program output, and are labelled as such: “PCAngsd” (the program no longer has the feature) and, in the first version of this comparison, “PC-Relate”. The latter was subsequently validated against the genuine `pcrelate()`: with `small.samp.correct = FALSE` the two agree at a correlation of 1.0000 and identical RMSE, which is what licenses using the re-implementation to stand in for PCAngsd. All kinship values are on the ϕ scale; evalAdmix output was halved to get there.

9 References

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